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INDEPENDENT APPLIED RESEARCH

Deepfake Image Detection Using a Hybrid CNN– ResNet Architecture

**with Evaluation-Driven Training and Dataset Integrity
Analysis**

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Research Domain: Deep Learning | CV

October, 2025

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Abstract

Deepfake generation technologies have rapidly advanced due to improvements in deep generative models such as GANs, autoencoders, and diffusion architectures. While these methods enable creative and beneficial use cases, they pose significant threats including misinformation, identity fraud, political manipulation, and cybercrime. This research presents an applied Deepfake image detection system using a Hybrid CNN–ResNet architecture trained on a large dataset of **141,083 real and synthetic face images** from public sources including CelebA, FaceForensics++, DFDC, HuggingFace, and Kaggle, along with custom-augmented variations.

The model was trained using transfer learning and a two-phase fine-tuning workflow with structured evaluation tracking and callback-driven optimization. During experimentation, unusually high validation accuracy and near-perfect metric curves led to an investigation into dataset handling. Analysis revealed **dataset leakage**, caused by mixing augmented copies with raw images prior to train-validation splitting, which allowed visually similar samples to appear across splits. This resulted in artificially inflated generalization performance. The dataset pipeline was redesigned to apply augmentation **only after splitting**, ensuring independence between training and validation data. Retraining and updated benchmark evaluation are currently in progress.

This work demonstrates the importance of rigorous evaluation, reproducibility, and scientific analysis in deepfake forensics, emphasizing that suspiciously high performance must be questioned rather than accepted at face value.

Keywords

Deepfake Detection, Hybrid CNN–ResNet, Transfer Learning, Dataset Leakage, Image Forensics, Deep Learning, Computer Vision, Evaluation Integrity

1. Introduction

Advancements in generative artificial intelligence have enabled the creation of highly realistic synthetic face images known as Deepfakes. While these technologies support entertainment and accessibility applications, malicious use cases such as political propaganda, identity manipulation, and personal harassment have raised major security concerns. Detecting Deepfakes reliably has therefore become an active research field in digital forensics and machine learning.

Traditional Deepfake detection strategies rely on handcrafted forensic cues such as eye blinking, blending boundaries, and texture inconsistencies. However, modern generation models significantly reduce such artifacts, limiting classical feature-based approaches. Deep learning-based detectors, particularly transfer-learning models and CNN architectures, have demonstrated the ability to learn subtle spatial inconsistencies from image pixel distributions.

This research presents a Hybrid CNN-ResNet architecture evaluated through structured training and validation. A core contribution of this work is the **analysis and correction of dataset leakage**, a frequently overlooked cause of misleading accuracy in deep learning literature. The project prioritizes reproducibility, scientific integrity, and the importance of evaluation-driven learning.

2. Related Work

Deepfake detection research includes:

CNN-based spatial detectors, using architectures like Xception, ResNet, VGG and EfficientNet.

Temporal-based models, incorporating recurrent or transformer architectures for video frame sequences.

Frequency-based forensic approaches, analyzing spectral artifacts left by generative models.

Hybrid models combining pretrained backbones and custom classifiers for improved feature representation.

Common challenges noted in literature include:

Poor robustness under cross-dataset testing,

Limited reproducibility

High dependency on dataset structure,

Risk of overfitting and leakage due to data processing errors.

This work contributes by:

Using a large multi-source dataset,

Performing evaluation-driven analysis,

Identifying and correcting leakage conditions rather than ignoring anomalies.

3. Dataset and Preprocessing

3.1 Dataset Composition

The dataset consists of **141,083 labeled images** split into balanced real and fake classes as follows:

Split	Real	Fake	Total
Train	56,217	56,213	112,430
Validation	3,514	3,514	7,164
Test	10,950	10,539	21,489

Sources included:

CelebA real face dataset

FaceForensics++

DFDC fake dataset

Kaggle and HuggingFace

Student-generated augmented/processed synthetic variations

3.2 Preprocessing

Resizing to $128 \times 128 \times 3$

Normalization

Directory-based dataset organization

Loading via `tf.keras.utils.image_dataset_from_directory`

Prefetching for optimized GPU utilization

3.3 Data Leakage Investigation

Earlier pipeline versions mixed augmented copies with original images *before splitting*, causing training and validation sets to share visually similar samples. This artificially inflated performance, as seen through smooth and near-perfect metric curves.

The updated pipeline splits raw images first, then applies augmentation only to training data.

4. Hybrid CNN–ResNet Architecture

The proposed model integrates a transfer-learning backbone with a custom CNN-based classification head. The architecture is implemented in a modular form for multi-phase fine-tuning.

4.1 Backbone Network

Model: ResNet50 (ImageNet pre-trained)

Configuration:

include_top=False for feature extraction

Frozen during Phase-1 training to preserve pretrained knowledge

4.2 Custom Classification Head

Using output feature maps from ResNet50:

Global Average Pooling (reduces parameters + combats overfitting)

Dense Layer (512 units, ReLU) — feature refinement

Batch Normalization — gradient stability

Dropout (0.3) — regularization

Dense Layer (128 units, ReLU) — feature compression

Batch Normalization and Dropout (0.2)

Final Dense Unit (1 neuron, Sigmoid) — binary classification

4.3 Input and Output

Parameter	Value
Input Size	$128 \times 128 \times 3$
Output	1 probability (Real vs Fake)

antic feature extraction while improving decision boundaries through custom trainable layers.

5. Training Strategy

Training was executed in two distinct phases using **the same architecture**:

Phase	Settings	Purpose
Initial Training	LR = 1e-4, 15 epochs, ResNet frozen	Warm-up classifier head
Fine-Tuning	LR = 1e-5, 5 epochs, last 20 ResNet layers unfrozen	Improve generalization

Additional training configuration:

Batch size = 20

Loss: Binary Crossentropy

Optimizer: Adam with gradient clipping (clipnorm=1.0)

Class weights computed for balanced learning

Callbacks:

ModelCheckpoint

EarlyStopping (patience=3)

ReduceLROnPlateau (factor=0.3)

Custom per-epoch graph logger

The model was evaluated using a held-out test split, maintaining full dataset independence.

6. Evaluation

Evaluation performed using the held-out test split.

Metrics recorded: accuracy and loss via `model.evaluate(test_ds)`.

Observed behavior:

Training and validation curves were unusually smooth

Very high performance metrics indicated unrealistically strong generalization

These results motivated deeper analysis of pipeline handling and data distribution.

7. Critical Analysis of Dataset Leakage

Inspection concluded that unusually high performance was likely attributable to:

Augmentation performed before splitting,

Visual similarity between samples across splits,

Validation and test sets not fully independent.

This demonstrates scientific integrity: **the result was questioned** rather than celebrated without verification.

Current status:

Clean pipeline built with split-first design

Retraining with corrected dataset is **pending**

Updated performance metrics will be analyzed for generalization

8. Deployment

A simple real-time inference application was developed using Streamlit, enabling:

Image upload

Model prediction

Visualization of output probabilities

Explainability features (e.g., Grad-CAM++) are planned for integration but **not yet implemented**.

9. Conclusion and Future Work

This project presents an applied Deepfake detection system using a hybrid CNN–ResNet model trained on a large dataset. Beyond architecture performance, the work emphasizes rigorous evaluation practices, reproducibility, and responsible interpretation of results. Identifying likely dataset leakage and redesigning the pipeline reflects a research-oriented thought process and professional engineering methodology.

Future directions include:

Retraining after leakage correction and publishing revised performance

Explainability integration (Grad-CAM++)

Transformer-based architectures

Spectral-domain Deepfake detection

Cross-dataset testing for real-world robustness